EVERY QUASI-COHERENT MODULE IS A DIRECTED COLIMIT OF KAPPA-GENERATED QUASI-COHERENT SUBMODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the choice of the covering data with a cardinal bounding all index sets, the refinement operator on families of subsets together with its iteration to a stable family and the glueing to a kappa-generated subsheaf, and the directedness and exhaustiveness of the resulting system.

Editorial note. The label lemma-colimit-kappa names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source properties.tex –label lemma-colimit-kappa –start 3771 –end 3848 –local-id 28039 returned RESULT=OK (L28039 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-properties/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Base change of sections of a quasi-coherent module along an inclusion of affine opens. Schemes, the lemma identifying the sections of the restriction of a quasi-coherent module with the tensor product of the sections over the smaller affine open. Supplies the identification M_i tensor over A_i with A_ijk = M_ijk = M_j tensor over A_j with A_ijk used to define the refinement map.
- Description of quasi-coherent modules on a scheme by their values on an affine open covering. Schemes, the section describing quasi-coherent modules through the values on the members of an affine open covering and the glueing data. Supplies the construction of a quasi-coherent subsheaf from a compatible family of submodules.

Source reference. Original lemma 01I9: sources/stacks-chow/tags.txt, source lines 1970–1970. Prerequisite pointer invoked at proof line(s) 3793; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let X be a scheme. There exists a cardinal κ such that every quasi-coherent module $\mathcal{F}$ is the directed colimit of its quasi-coherent κ-generated submodules.*

Proof. Choose an affine open covering $X = \bigcup_{i \in I} U_i$. For each pair i, j choose an affine open covering $U_i \cap U_j = \bigcup_{k \in I_{ij}} U_{ijk}$. Write $U_i = \mathrm{Spec}(A_i)$ and $U_{ijk} = \mathrm{Spec}(A_{ijk})$. Let κ be any infinite cardinal $\geq$ than the cardinality of any of the sets I, I_{ij}.

Let $\mathcal{F}$ be a quasi-coherent sheaf. Set $M_i = \mathcal{F}(U_i)$ and $M_{ijk} = \mathcal{F}(U_{ijk})$. Note that

$$M_i \otimes_{A_i} A_{ijk} = M_{ijk} = M_j \otimes_{A_j} A_{ijk}.$$

see Schemes, Lemma 01I9. Using the axiom of choice we choose a map

$$(i, j, k, m) \mapsto S(i, j, k, m)$$

which associates to every $i, j \in I$, $k \in I_{ij}$ and $m \in M_i$ a finite subset $S(i, j, k, m) \subset M_j$ such that we have

$$m \otimes 1 = \sum\nolimits_{m' \in S(i,j,k,m)} m' \otimes a_{m'}$$

in M_{ijk} for some $a_{m'} \in A_{ijk}$. Moreover, let's agree that $S(i, i, k, m) = \{m\}$ for all $i, j = i, k, m$ as above. Fix such a map.

Given a family $\mathcal{S} = (S_i)_{i \in I}$ of subsets $S_i \subset M_i$ of cardinality at most κ we set $\mathcal{S}' = (S_i')$ where

$$S_j' = \bigcup\nolimits_{(i,k,m) \text{ such that } m \in S_i} S(i, j, k, m)$$

Note that $S_i \subset S_i'$. Note that S_i' has cardinality at most κ because it is a union over a set of cardinality at most κ of finite sets. Set $\mathcal{S}^{(0)} = \mathcal{S}$, $\mathcal{S}^{(1)} = \mathcal{S}'$ and by induction $\mathcal{S}^{(n+1)} = (\mathcal{S}^{(n)})'$. Then set $\mathcal{S}^{(\infty)} = \bigcup_{n \geq 0} \mathcal{S}^{(n)}$. Writing $\mathcal{S}^{(\infty)} = (S_i^{(\infty)})$ we see that for any element $m \in S_i^{(\infty)}$ the image of m in M_{ijk} can be written as a finite sum $\sum m' \otimes a_{m'}$ with $m' \in S_j^{(\infty)}$. In this way we see that setting

$$N_i = A_i\text{-submodule of } M_i \text{ generated by } S_i^{(\infty)}$$

we have

$$N_i \otimes_{A_i} A_{ijk} = N_j \otimes_{A_j} A_{ijk}.$$

as submodules of M_{ijk}. Thus there exists a quasi-coherent subsheaf $\mathcal{G} \subset \mathcal{F}$ with $\mathcal{G}(U_i) = N_i$. Moreover, by construction the sheaf $\mathcal{G}$ is κ-generated.

Let $\{\mathcal{G}_t\}_{t \in T}$ be the set of κ-generated quasi-coherent subsheaves. If $t, t' \in T$ then $\mathcal{G}_t + \mathcal{G}_{t'}$ is also a κ-generated quasi-coherent subsheaf as it is the image of the map $\mathcal{G}_t \oplus \mathcal{G}_{t'} \to \mathcal{F}$. Hence the system (ordered by inclusion) is directed. The arguments above show that every section of $\mathcal{F}$ over U_i is in one of the $\mathcal{G}_t$ (because we can start with $\mathcal{S}$ such that the given section is an element of S_i). Hence $\mathrm{colim}_t \mathcal{G}_t \to \mathcal{F}$ is both injective and surjective as desired. $\square$